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Sloshing analysis and mechanical modeling of liquid cargo during lifting

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Abstract

When the acceleration of the liquid-filled container undergoes a step change, liquid sloshing is prone to occur. A sloshing linear dynamic model is put forward to predict the position change of center of mass of sloshing liquid. This modeling process uses a liquid static model and adds a time variable. The numerical simulation example finally verifies the feasibility of the mathematical model and shows that linear dynamic model has the ability to predict the displacement of the center of mass.

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1. Introduction

The prediction of liquid sloshing characteristics is very necessary in various fields. In the process of moving molten metal in the metallurgical industry, the liquid level in the ladle is an unconstrained boundary condition, and it is difficult to control the magnitude of the excitation imposed on the tank. Thus, the change of the position of the center of mass caused by the liquid sloshing is prone to occur, which increases the probability of tank tipping. For this reason, people need to predict the changing law of the center of mass in the case of tank shaking, so as to improve the controllability of the liquid-filled tank during the transportation. Therefore, there is an urgent need for a mathematical model that can accurately predict the change in the center of mass of the sloshing liquid.

Present scholars mainly use analytical methods or numerical methods to analyze the sloshing characteristics. In the study of weak nonlinear sloshing, linear sloshing theory is generally used^[1]. It can predict the dynamic

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characteristics of liquid sloshing more accurately than the equivalent mechanical model when the premise of small sloshing and excitation frequency far from the resonance frequency. Under conditions of stronger nonlinearity, it is mainly to use nonlinear fluid vibration theory^[2] to analyse or use model theory^[3] to superimpose multiple modes to solve the problem. Numerical calculation methods that are widely recognized and used include computational fluid dynamics(CFD)^[4] methods, Marker-And-Cell Method(MAC)^[5], Volume of fluid(VOF)^[6], Finite Volume Fluid^[7]. However, analytical methods need to be analyzed based on specific premises, and numerical calculation methods require high calculation costs. For this reason, people often use static models at equilibrium moments to study liquid sloshing in order to reduce calculation costs. For example, Wei^[8] simplified their fluid models to static models to analyze the approximate calculation of the center of gravity of furnace liquid dumping. The research only considers the constraints of the container and the acceleration of gravity and to focus on the relationship between the inclination angle of the container and the center of gravity. Huang Hua^[9] used the centroid surface equivalent model to study the phenomenon of large-scale fluid sloshing and the force of the liquid sloshing on the tank is finally obtained.

In this article, we try to propose a sloshing linear dynamic model. This modeling method uses a static model to describe the state of fluid equilibrium. It regards the change of force on the liquid system as linear and assumes the liquid level is flat. According to this modeling method, the linear dynamic equation is derived, and the displacement of the liquid center of mass is solved. The validity of this modeling method and the accuracy of the composite model are verified through numerical examples. The numerical calculation software FLUENT finally verified the feasibility of the mathematical model.

2. Modeling method

Supposing the liquid cargo moves in a plane. Figure 1 is the static liquid model at the initial moment. And Figure 2 is the static liquid model at the equilibrium moment. Oxz is the coordinate system that moves with the liquid system. The tank moves in the x direction with a constant acceleration of a_t . The acceleration of the liquid system in the x axis direction is a_0 . l is the length of the liquid model along the x axis. w is the length of the liquid model along the y axis. h is the undisturbed static liquid surface. g is the acceleration of gravity. θ is the angle between the liquid surface and the horizontal plane. $h+z_l$ is the contact surface between the left container wall and the liquid at the time of equilibrium. $h-z_l$ is the contact surface between the right container wall and the liquid at the moment of equilibrium. ρ is the density of the liquid system.

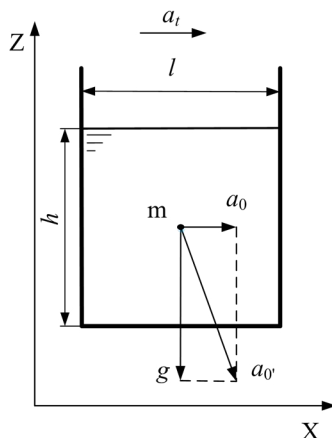


Fig. 1. Static liquid model at the initial moment

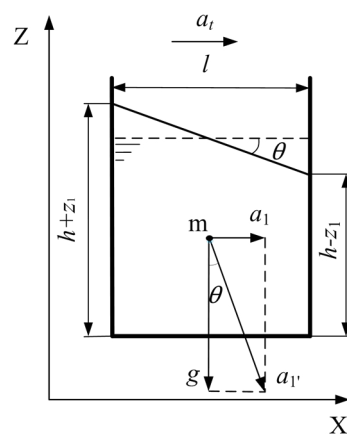


Fig. 2. Static liquid model at the equilibrium moment

At the initial moment, the pressure difference in the x -axis direction forms the force of the fluid system in the x -direction.

$$p_0 = \rho a_0 l \quad (1)$$

The force of the fluid along the x axis can be expressed as

$$F_0 = p_0 wh = \rho lwha_0 \quad (2)$$

At the moment of equilibrium, the force along the x axis is the sum of the force on the shaking part and the force on the relatively static part $F_1 = F_{11} + F_{12}$. The force on the relatively static part along the x axis can be expressed as

$$F_{12} = \rho lw(h - l \tan \theta) a_1 \quad (3)$$

In this formula, a_1 is the acceleration of the fluid at the moment of equilibrium. When the liquid system reaches equilibrium, $a_1 = a$. The relative shaking part can be expressed as

$$F_{11} = \rho w a_1 \int_0^{l \tan \theta} \frac{l \tan \theta - z}{\tan \theta} dz \quad (4)$$

Then the force of the entire system along the x axis can be finally simplified as

$$F_1 = \rho l w a_1 \left(h - \frac{3 l \tan \theta}{2} \right) \quad (5)$$

The effect of the external force on the system is the distance of center of mass of the sloshing liquid in the container moves, and the distance L of the external force is expressed as the following.

$$L = \frac{a_1 t^2}{2} - \frac{l}{2} + \frac{l^2}{12h} \tan \theta \quad (6)$$

The potential energy of the system is the gravitational potential energy of the liquid.

$$E_p = \frac{l \tan \theta}{12} \rho g l w \frac{l \tan \theta}{2} \quad (7)$$

Since only the motion in the xz plane is considered, the kinetic energy of the liquid system at equilibrium can be expressed as

$$E_k = \frac{mv^2}{2} \quad (8)$$

Considering that the change of differential pressure resistance before and after stabilization is extremely small, the change of differential pressure resistance during the change process can be regarded as a linear change, and the motion of the system satisfies the law of conservation of energy. Under the premise of ignoring the influence of turbulence, the energy conservation equation is introduced to obtain.

$$\frac{l}{4} \left(a_1 t^2 - l + \frac{l^2 a_1}{6 h g} \right) \left[\rho l w \left(h a_0 + h a_1 - \frac{3 a_1^2 l}{2 g} \right) \right] = \frac{\rho l^3 w a_1^2}{24 g} + \frac{\rho l w h (a_1 t)^2}{2} \quad (9)$$

ΔL_X is the difference between the displacement of the center of gravity of the container and the center of gravity of the fluid along the x axis at the moment of equilibrium, and can also be regarded as the difference between the half of the length of the fluid model along the x axis and the distance between the right-angle trapezoidal center of gravity formula and the long side.

$$\Delta L_X = \frac{l^2 a_1}{12 h g} = \frac{(a_1 - a_0)}{2} t^2 \quad (10)$$

According to equation (10), t can be transformed into the following form.

$$t^2 = \frac{a_1 l^2}{6 h g (a_1 - a_0)} \quad (11)$$

From the known quantities a_1 , h , g , l , w , ρ , the offset of the center of gravity along the z axis at the time of balance and the offset of the center of gravity along the x axis at the time of balance can be calculated.

$$\begin{cases} x_1 = -\frac{l^2 a_1}{12 h g} \\ z_1 = \frac{l^2 a_1^2}{24 h g^2} \end{cases} \quad (12)$$

According to formulas(9) and (11), the time it takes for the system to reach equilibrium can be obtained

$$\begin{aligned} t = & (a_1 \div (((1 \cdot ((l^{12} a_1^4 - 12 l^{10} a_1^4 g h^2 - 216 l^9 a_1^5 g h^3 - \\ & 144 l^9 a_1^3 g^2 h^2 + 180 l^8 a_1^4 g^2 h^4 + 432 l^7 a_1^3 g^3 h^4 + \\ & 6912 l^6 a_1^4 g^3 h^5 + 7776 l^6 a_1^2 g^4 h^4 - 2592 l^5 a_1^3 g^4 h^6 + \\ & 15552 l^4 a_1^2 g^5 h^6 + 31104 l^3 a_1^3 g^5 h^7 - 186624 l^3 a_1 g^6 h^6 \\ & + 46656 l^2 a_1^2 g^6 h^8 - 559872 l a_1 g^7 h^8 + 1679616 g^8 h^8)^{1/2} + \\ & l^6 a_1^2 + 1296 g^4 h^4 + 216 l a_1 g^3 h^4 - 72 l^3 a_1 g^2 h^2 - 18 l^4 a_1^2 g h^2)) \\ & \div (-72 a_1 l^3 g^2 h^3 + 2592 g^4 h^5)) - a_1))^{1/2} \end{aligned} \quad (13)$$

Then according to the offset of the center of gravity when the x and z axes are balanced, the acquired dynamic center of gravity coordinates are finally obtained

$$\begin{cases} x_i = -\frac{l^2 a_i t_i}{12 h g t} \\ z_i = \frac{l^2 a_i^2 t_i}{24 h g^2 t} \end{cases} \quad (14)$$

3. Simulation and experiment

Taking a rectangular container partially filled with liquid as an example, the calculation results of the fluid sloshing linear dynamic model at the position of the liquid mass center under translational excitation are compared with the simulation results of the CFD software Fluent and the calculation results of the traditional static model to verify linear dynamics feasibility of modeling method. The liquid used in the calculation example is water at room temperature, with a density $\rho=998.2\text{kg/m}^3$ and a dynamic viscosity coefficient $\mu=0.00100\text{Ns/m}^3$. The bottom surface of the rectangular container is a square with a side length of 400mm and a height of 600mm. To ensure that the residuals of the two-phase flow model can be calculated smoothly. The grid is set to a regular hexagon, and its network division and grid accuracy settings are shown in Table 1.

Table 1. Grid Accuracy

Minimum solid mesh size	Maximum surface mesh size	Maximum solid mesh size
10^{-3}	2×10^{-3}	2×10^{-3}

In order to verify the effectiveness and accuracy of linear dynamic model of liquid sloshing proposed in this paper under translational excitation, external excitations with the same speed of 0.65 m/s^2 is applied to make the container move along the x axis and y axis. The displacement of the fluid centroid position of the sloshing linear dynamic model on the x axis and y axis is compared with the CFD simulation results and the calculation results of the traditional static model. The comparison results of the centroid offset on the x axis and the y axis are shown in Figure 3, Figure 4.

Among them, the calculation results of the static model and the sloshing linear dynamic model are consistent. At zero time of acceleration input (initial time), the position of the center of mass has reached the position of the center of mass in the equilibrium state, there is no transition process in between and the offset of the centroid coordinate on the x axis and y axis is constant. Then the linear dynamic model's barycentric coordinate offset curve along the two axes is a horizontal broken line after rising with a constant slope and reaching a stable output value. The center of gravity offset curves of the two eventually coincide.

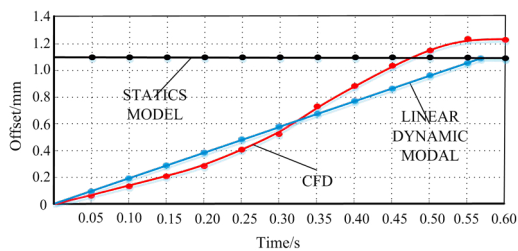


Fig. 3. The offset curve of the center of mass coordinate in the X-axis direction under acceleration 0.65 m/s^2

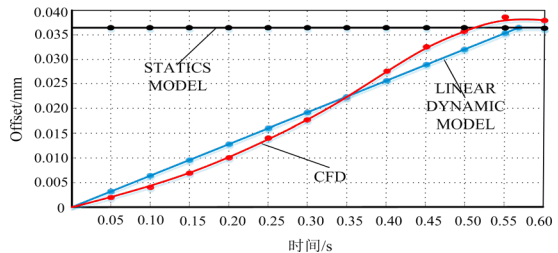


Fig. 4. The offset curve of the center of mass coordinate in the Y-axis direction under acceleration 0.65 m/s^2

In the comparison between the linear dynamic model and CFD at acceleration of 0.65 m/s^2 , the CFD simulation model in the x and y axis of center of gravity coordinate offset curve shows a slope that first increases and then decreases. When the curve of CFD reaches its peak, it begins to fall back, and then its simulation model will

oscillate up and down until it reaches a stable output state. It can be seen that in the process of fluid from static to equilibrium, the calculation of the linear dynamic model and the centroid position offset of CFD shows great consistency, and the change trend is consistent. Compared with the step change process of the centroid position of the traditional static model, the linear dynamic model is greatly improved.

4. Conclusion

Liquid responses under constant translational acceleration illustrate that liquid system has a continuous change process between the static state and the static model. Thus, Sloshing linear dynamic model is proposed and fluid sloshing process is assumed to a linear process. The position of the center of mass in the equilibrium state is determined by the liquid level equation. For rectangular storage box, a time-domain equation for calculating the change of the center of mass is proposed. And In this equation, there is no need to use CFD software to calculate the structural parameters in the model. Verification of the modeling method is conducted by comparison with CFD software. Experiments show that the linear dynamic model and the experimental results of CFD calculation are in good agreement.

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